BIT 4131 Statistical Computing and Reporting

Answer all Questions

Acknowledge use of AI/LLM if any and also indicate your own critical thinking. Put references and citations where necessary. APA 6th edition

QUESTION ONE (20 MARKS)

The clinic network models arrivals for staffing decisions and estimates uncertainty in waiting times. At one small clinic, daily arrivals are modelled as Poisson with mean 3 per day.

1. Explain why increasing the sample size reduces the variation of sample means under independent sampling. (2 marks)
2. Write Python or R code to calculate $P(X = 0)$ and $P(X \geq 1)$ (2 marks). Using a fixed seed, draw 1,000 independent samples of 30 daily counts, store their means, plot their distribution and calculate their empirical standard error
3. Observed daily counts have mean 3 and variance 12. Analyse the departure from the Poisson model and identify one possible explanation and one data check.
4. Design a bootstrap study to estimate uncertainty in the mean of waiting times = [8, 10, 12, 15, 18, 20, 25, 40]. Justify the sampling unit, and provide Python or R code for 1,000 bootstrap means and a percentile confidence interval.

QUESTION TWO (20 MARKS)

For service planning, daily arrivals x = [10, 20, 30, 40] and corresponding mean waiting times y = [12, 16, 19, 23] minutes are modelled. The data frame patients contains Age, WaitingMinutes and VisitsPerYear. The covariance matrix in part (b) describes two numerical patient-profile variables.

1. Explain the role of the intercept and the meaning of a fitted waiting time in the clinic's regression model.
2. Write Python or R code to fit an intercept-and-slope least-squares model and predict mean waiting time when daily arrivals x = 50
3. A regression residual plot has a pronounced curved pattern. Analyse what this suggests about the model and propose a modification, including one way to assess the improvement.
4. Design a principal component analysis workflow in Python or R for Age, WaitingMinutes and VisitsPerYear in patients.
5. The network models high satisfaction as $\log(p / (1 - p)) = -1 + 0.4x$, where p is the probability of a satisfaction rating of at least 4 on a five-point scale and x is visits per year. A report says each extra visit adds 0.4 directly to probability. Evaluate this statement and show how to calculate the probability and odds ratio correctly.

QUESTION THREE (20 MARKS)

The network forecasts attendance using a data frame attendance with Month and Visits for 36 consecutive months. A simplified forecast-calibration cost is $g(\theta) = (\theta + 2)^2 + 5$, where θ is an additive forecast adjustment that may be positive or negative.

1. Explain autocorrelation and its relevance to time-series forecasting.
2. Write Python or R code to plot monthly visits and calculate month-to-month differences
3. Missing attendance values cluster during a service disruption. Analyse why deleting those months may bias the report,
4. The clinic attendance forecast uses information from future months when constructing predictors and omits its prediction interval. Evaluate the effect on reported performance and the usefulness of the forecast for planning.

QUESTION FOUR (20 MARKS)

A clinic network evaluates service performance to reduce waiting times and plan staffing. Anonymised records contain PatientID, Clinic, Age, WaitingMinutes and SatisfactionRating. Questions One to Five use different samples and models from this study; each question supplies the data needed to answer it.

1. Discuss why the median may be preferred to the mean for highly skewed waiting times.
2. Explain why synthetic patient records used in the service study must be labelled and their generation documented.
3. Write Python or R code to create a data frame with Clinic = [A, A, B, B] and WaitingMinutes = [10, 20, 15, 25].
4. For waiting times = [10, 12, 15, 18, 45], write Python or R code to calculate the median, sample standard deviation and interquartile range.
5. Using the data frame in part (i), write Python or R code to calculate the mean waiting time for each clinic.

QUESTION FIVE (20 MARKS)

The network evaluates a new appointment system. Waiting times in minutes for the same six patients are Before = [25, 30, 28, 35, 32, 26] and After = [20, 24, 23, 27, 26, 21]. It also compares attendance patterns and waiting times across clinics. Use a 5% significance level.

1. Explain the null and two-sided alternative hypotheses for the mean within-participant change.
2. Write Python or R code to perform a paired t-test and display its statistic and p-value
3. . A clinic expects equal attendance on three days but observes [45, 30, 15]. Perform a chi-square goodness-of-fit test and display expected counts (3 marks).

4. The six patients have ages [20, 35, 30, 45, 40, 25], paired in order with the Before waiting times. Write Python or R code to calculate sample covariance and Pearson correlation between age and waiting time, then repeat after converting minutes to seconds. Analyse why covariance changes while correlation remains unchanged.
5. Evaluate whether an independent-samples test is appropriate for the Before and After data, and justify your judgement using the study design.
6. Paired differences are strongly skewed and contain an extreme value. Select a suitable paired non-parametric alternative, justify your choice and state one assumption.